

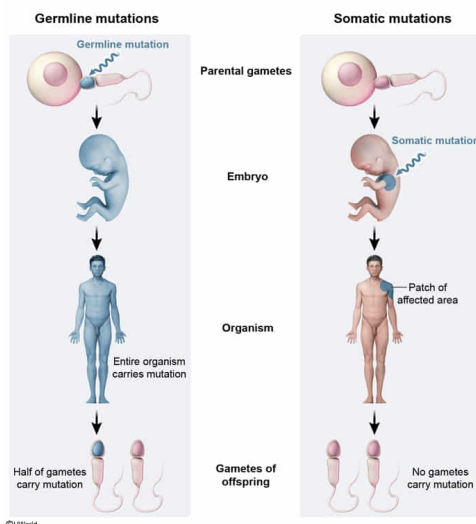
Assume L410 is an acquired mutation that occurs only in hepatocytes of adults who were exposed to a certain mutagen. A woman with the L410 mutation has a child with an unaffected man. What is the probability that this child has the L410 mutation?

- ✓ ☒ A. 0.00 [77%]
☐ B. 0.25 [12%]
☐ C. 0.75 [2%]
☐ D. 1.00 [6%]

Correct

78% answered correctly

Explanation:



All cells in the body can be classified as either **germ** (reproductive cells) or **somatic** (ie, the remaining cells). Germ cells are progenitor cells that undergo **meiosis** to produce gametes, such as sperm and oocytes. **Germline mutations**, which occur in the germ cells of parents, can be **passed to offspring** during sexual reproduction because parental gametic cells combine to form a **zygote** (fertilized egg). In contrast, parental **somatic mutations**, which involve alteration of somatic cell lineage (DNA) *after* conception, **will not pass to an offspring** because somatic cells are not directly involved in zygote formation.

According to the question, the *acquired* mutation, L410, occurs only in the hepatocytes (liver cells) of adults due to exposure to a mutagen (agent causing genetic mutation). This means that the L410 mutation is **somatic** because it occurred sporadically in a minority of cells and *not* in the cells of the entire organism (as would be seen in germline mutations). It can be concluded that L410 occurred during the lifetime of the individual and *not* in the gametes of his or her parents. Therefore, the somatic **L410 mutation** in the mother will **not be inherited** by the offspring, and the probability of this woman and a genetically normal man having a child with the L410 mutation is equal to **zero (Choices B, C, and D)**.

Educational objective:

Mutations that occur in the reproductive cells of parents are called germline mutations. These mutations can be passed from parent to offspring, and result in the offspring carrying the mutation in all of its cells. Parental somatic mutations, which occur in non–sex cell lineages during an organisms' lifetime, are not inherited by the offspring.

Time spent: 0 sec
Subject:
Biology

QID: 401518
System:
Molecular Biology